

Total

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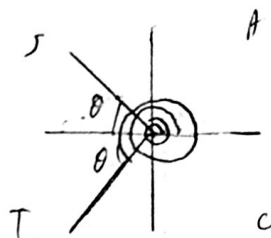
1. Solve for x on the interval $0 \leq x \leq 2\pi$. Give the exact solution(s) in radian measure, where possible. Otherwise, round to the nearest hundredth of a radian.

a) $2\cos\left(2x - \frac{\pi}{2}\right) = -\sqrt{2}$

$$2x - \frac{\pi}{2}$$

$$= 2\left(x - \frac{\pi}{4}\right)$$

$$= 2\theta$$



$$\cos\left(2x - \frac{\pi}{2}\right) = \frac{-\sqrt{2}}{2}$$

$$\cos \theta = -\frac{\sqrt{2}}{2}$$

$$\theta = \frac{\pi}{4}$$

$$2x - \frac{\pi}{2} = \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{11\pi}{4}, \frac{13\pi}{4}$$

$$x = \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}$$

b) $-6\cos x = 3\tan x$

$$-2\cos x = \frac{\sin x}{\cos x}$$

$$-2\cos^2 x = \sin x$$

$$-2(1 - \sin^2 x) = \sin x$$

$$-2 + 2\sin^2 x = \sin x$$

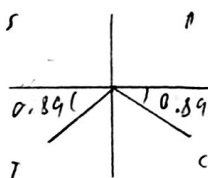
$$2\sin^2 x - \sin x - 2 = 0$$

$$\sin x = \frac{1 \pm \sqrt{(-1)^2 - 4(2)(-2)}}{2(2)}$$

$$\sin x = -0.78 \text{ or } 1.28$$

DNE

$$\sin x = -0.78$$



$$x = 4.03, 5.39$$

2. Give an equation for a sine function that has a minimum at $\left(\frac{5\pi}{6}, 10\right)$, followed immediately by another minimum at $\left(\frac{11\pi}{6}, 10\right)$, and has a maximum value of 22. (fill in the blanks)

$$y = 6 \sin\left[2\left(x - \frac{13\pi}{12}\right)\right] + 16$$

$$a = \frac{22-10}{2} = 6 \quad p = \frac{11\pi}{6} - \frac{5\pi}{6} = \pi$$

$$c = \frac{21\pi/6}{2} = 16 \quad k = \frac{2\pi}{\pi} = 2$$

$$\frac{p}{4} = \frac{\pi}{4} \quad \frac{\pi}{4} + \frac{5\pi}{6} = \frac{13\pi}{12}$$

sin x starts middle of min & max

3. Determine the average rate of change of the function $f(x) = \sin x$ from $x = \frac{5\pi}{6}$ to $x = \frac{11\pi}{6}$.

Round to 2d.p.

$$\left(\frac{5\pi}{6}, \frac{1}{2}\right) \quad \text{AROC} = \frac{-\frac{1}{2} - \frac{1}{2}}{\frac{11\pi}{6} - \frac{5\pi}{6}} = -0.32$$

$$\left(\frac{11\pi}{6}, -\frac{1}{2}\right)$$

4. Jovana likes the new roller coaster at Canada's Wonderland. Instead of riding it, Jovana decided to come up with the sinusoidal equation that best models the coaster (she loves math) !!! This is what she came up with.

$$H(t) = 150 \sin\left(\frac{\pi}{30}t\right) + 250 \text{ where } H(t) \text{ is height in meters and } t \text{ is time in seconds.}$$

- a) When does the coaster reach its minimum height?

$$\sin\left(\frac{\pi}{30}t\right) = -1$$

$$\frac{\pi}{30}t = \frac{3\pi}{2}$$

$t = 45$ seconds and every 60 seconds after

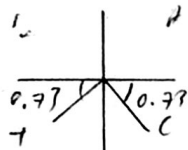
- b) How long is the coaster above 150m over the first 100 seconds?
(to the nearest second)

$$150 = 150 \sin\left(\frac{\pi}{30}t\right) + 250$$

$$-\frac{2}{3} = \sin\left(\frac{\pi}{30}t\right)$$

$$\sin \theta = -\frac{2}{3}$$

$$p = \frac{2\pi}{\frac{\pi}{30}} = 60$$



$$\frac{\pi}{30}t = 3.87, 5.55, 10.15, 11.83 \text{ net range.}$$

$$t = 37, 53, 97$$

↑ ↑ ↑
1st time 1st time 3rd time
going up going up going down
under 150m

$$37 + (97 - 53) = 81 \text{ seconds, above 150m}$$

For the first 100 seconds

5. Ursula is solving a trigonometric equation where

- a) What was her mistake? Explain why her reasoning $0 \leq x \leq 2\pi$ is incorrect.

(1) Her mistake was dividing $\cos x$ on both sides, which removes possible solutions.

$$3\cos^2 x = 2\cos x$$

$$3\cos x = 2$$

$$\cos x = \frac{2}{3}$$

$$x = 0.84 \text{ in quadrant 1}$$

$$\text{and } x = 5.44 \text{ in quadrant 4}$$

b) Give the correct solution.

$$3\cos^2 x - 2\cos x = 0$$

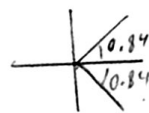
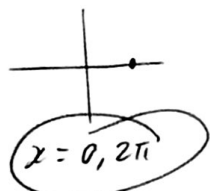
$$\cos x = 0$$

$$3\cos x - 2 = 0$$

$$\cos x = \frac{2}{3}$$

$$\cos x (3\cos x - 2) = 0$$

~~AF~~



$$x = 0.84, 5.44$$

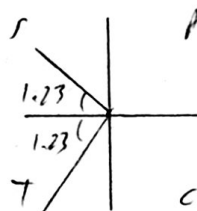
6. Solve for x on the interval $0 \leq x \leq 2\pi$. Give the exact solution(s) in radian measure, where possible. Otherwise, round to the nearest hundredth of a radian.

a) $[3\cos^2(x-1) - 3\sin^2(x-1)] = -1$

$$\cos 2\theta = \frac{-1}{3}$$

$$3[\cos^2(x-1) - \sin^2(x-1)] = -1$$

$$\cos[2(x-1)] = \frac{-1}{3}$$



$$2(x-1) = 1.91, 4.37, 8.19, 10.65, \text{ check } -\pi + 1.23 = -1.91 \text{ because of phase shift right}$$

$$x = 1.96, 3.18, 5.10, 6.33, 0.04$$

b) $\frac{1 + \sec x}{\tan x + \sin x} = \sqrt{2}$

$$1 + \sec x = \sqrt{2}(\tan x + \sin x)$$

$$1 + \frac{1}{\cos x} = \sqrt{2}\left(\frac{\sin x}{\cos x} + \sin x\right)$$

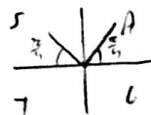
$$\frac{\cos x + 1}{\cos x} = \sqrt{2}\left(\frac{\sin x + \sin x \cos x}{\cos x}\right)$$

$$0 = \sqrt{2}\left(\frac{\sin x(1 + \cos x)}{\cos x}\right) - \frac{\cos x + 1}{\cos x}$$

$$0 = \left(\sqrt{2}\sin x - 1\right)\left(\frac{\cos x + 1}{\cos x}\right)$$

$$\sqrt{2}\sin x - 1 = 0$$

$$\sin x = \frac{1}{\sqrt{2}}$$

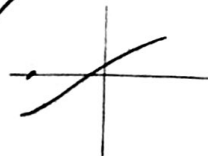


$$x = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\frac{\cos x + 1}{\cos x} = 0$$

$$\cos x + 1 = 0$$

$$\cos x = -1$$



$$x = \pi$$

$$\begin{aligned} \tan x + \sec x &\neq 0 \\ \sin x + \frac{1}{\cos x} &\neq 0 \\ x &\neq 0, \pi \end{aligned}$$

7. A cosine function has half the period of $y = \cos x$. All other parameters are the same between the two functions. Determine the points of intersection(s) if the two functions are graphed on the interval $x \in [0, 2\pi]$.

$$y = \cos x$$

$$p = \pi$$

$$k = 1$$

$$k = \frac{2\pi}{\pi} = 2$$

$$p = 2\pi$$

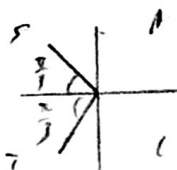
$$y = \cos 2x$$

$$\cos x = \cos 2x$$

$$\cos x = 2\cos^2 x - 1$$

$$0 = 2\cos^2 x - \cos x - 1$$

$$0 = (2\cos x + 1)(\cos x - 1)$$



$$2\cos x + 1 = 0$$

$$\cos x = -\frac{1}{2}$$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\cos x - 1 = 0$$

$$\cos x = 1$$

$$x = 0, 2\pi$$



8. Solve for x on the interval $0 \leq x \leq 2\pi$. Give the exact solution(s) in radian measure if possible.

a) $\cos\left(\frac{\pi}{4} + x\right) - \cos\left(\frac{\pi}{4} - x\right) = -1$

$$\cos\frac{\pi}{4}\cos x - \sin\frac{\pi}{4}\sin x - (\cos\frac{\pi}{4}\cos x - \sin\frac{\pi}{4}\sin x) = -1$$

$$-2\sin\frac{\pi}{4}\sin x = -1$$

$$2\left(\frac{\sqrt{2}}{2}\right)\sin x = 1$$

$$\sin x = \frac{1}{\sqrt{2}}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}$$

